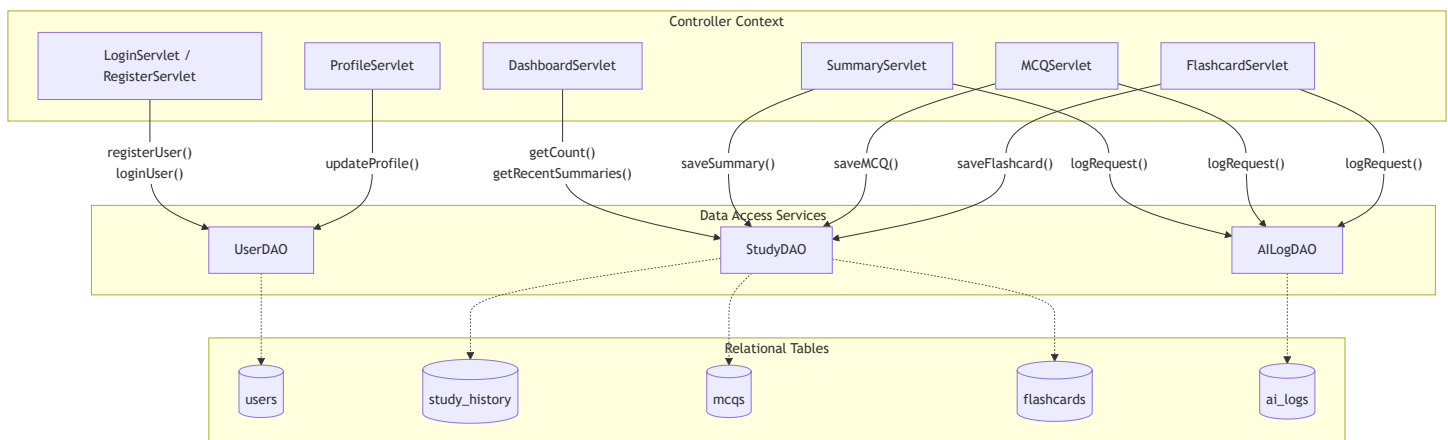


Component Interactions & Controller-DAO Mapping

This document details the internal module interaction topology of the **AI Study Companion** platform. It demonstrates how HTTP controllers interface with dedicated storage access units, infrastructure helpers, and third-party web endpoints.

1. Controller to DAO Dependency Graph

The platform decouples request mapping from structural execution. Controllers hold references to stateless Data Access Objects (DAOs) initialized during early servlet engine instantiation phases (`init()` callbacks).



2. Infrastructure Flow Mapping

2.1 File Extraction Layer (`FileExtractor.java`)

Invoked dynamically across study endpoints when client requests present `multipart/form-data` MIME declarations containing active binary input streams.

- **Consumers:** `SummaryServlet` , `MCQServlet` , `FlashcardServlet`
- **Behavior:**

- Inspects submitted file extension headers.
- If `.txt` is found: Instantiates standard Java stream buffering layers utilizing `UTF-8` character boundaries to rebuild text content.
- If `.pdf` is found: Passes streaming bytes into **Apache PDFBox's** `PDDocument` unzipping structure, executing `PDTextStripper` to strip text layouts cleanly.

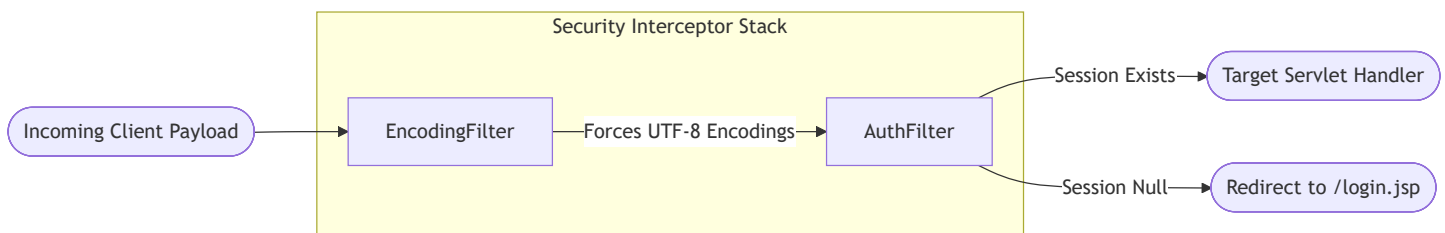
2.2 External AI Connector (`GeminiAPI.java`)

Centralized communication wrapper wrapping HTTP connectivity layer constraints.

- **Consumers:** `SummaryServlet` , `MCQServlet` , `FlashcardServlet`
- **Behavior:**
 - Pulls static external configurations (`AIza...` API keys) directly from `config.properties` resource bundles.
 - Constructs valid JSON request blocks encapsulating text contexts.
 - Establishes secure, outgoing REST API HTTP connections targeting Google Gemini public engine points (`https://generativelanguage.googleapis.com/...`).
 - Extracts internal generative candidates natively prior to string hand-offs.

3. Filter Execution Matrix

Before any mapped controller path executes, traffic routes through pre-configured filtering wrappers ensuring uniform serialization and access security.



3.1 EncodingFilter

- **Target Mapping:** `/*` (Global context traversal)
- **Role:** Resolves character dropouts caused by parsing dynamic multi-byte character strings, ensuring all `request.setCharacterEncoding("UTF-8")` properties bind cleanly.

3.2 AuthFilter

- **Target Mapping:** /dashboard , /summary , /mcq , /flashcards , /profile
- **Role:** Validates runtime server sessions, ensuring unauthenticated external traffic cannot bypass root authorization constraints to read internal study material endpoints directly.